

Systems of First Order Linear ODEs

1 First order ODE systems

The general system of first order equations involving n dependent variables defined on an interval $I \subseteq \mathbb{R}$ can be written as

$$\begin{cases} y_1'(t) = F_1(t, y_1, \dots, y_n) \\ y_2'(t) = F_2(t, y_1, \dots, y_n) \\ \vdots \\ y_n'(t) = F_n(t, y_1, \dots, y_n) \end{cases} \quad \text{for } t \in I. \quad (1)$$

with initial conditions $y_1(t_0) = x_1, \quad y_2(t_0) = x_2, \quad \dots, \quad y_n(t_0) = x_n,$

The solution to the system can be expressed in vector form as

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}$$

Theorem 1.1 (Existence and Uniqueness).

For the system of first order ODEs (1), if the functions $F_1, F_2, \dots, F_n$ and the partial derivatives $\frac{\partial F_i}{\partial y_j}$ for all pairs $i, j = 1, 2, \dots, n$ are continuous on a region R defined by

$$R := [t_0 - a, t_0 + a] \times [x_1 - b, x_1 + b] \times \dots \times [x_n - b, x_n + b] \subseteq \mathbb{R}^{n+1},$$

where constants $a, b > 0$, then there exists a unique solution $\mathbf{y}(t)$ for $t \in (t_0 - h, t_0 + h)$ to the IVP, where

$$h = \min\{a, b/M\} \quad \text{and} \quad M = \max_{\mathbf{v} \in R} \{|F_1(\mathbf{v})|, |F_2(\mathbf{v})|, \dots, |F_n(\mathbf{v})|\}.$$

Any n -th order ODE can be rewritten as a system of n first order ODEs. Given an n -th order ODE

$$y^{(n)} = F(t, y, y', \dots, y^{(n-1)}),$$

setting $z_1 = y, z_2 = y', \dots, z_n = y^{(n-1)}$, we obtain the system

$$\begin{cases} z_1' = z_2 \\ z_2' = z_3 \\ \vdots \\ z_{n-1}' = z_n \\ z_n' = F(t, z_1, z_2, \dots, z_n) \end{cases}$$

which is a system of n first order ODEs.

Definition 1.1 (First-Order Linear System).

A system of first order ODEs is called **linear** if it can be expressed in the form

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t) \quad \text{for } t \in I, \quad (2)$$

where

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} \quad \mathbf{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix} \quad \mathbf{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \cdots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \cdots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \cdots & p_{nn}(t) \end{pmatrix}$$

Definition 1.2 (Homogeneous First-Order Linear System).

A first order linear system is called **homogeneous** if $\mathbf{g}(t) = \mathbf{0}$ for all $t \in I$, i.e., it can be expressed as

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) \quad \text{for } t \in I.$$

Theorem 1.2 (Existence and Uniqueness for First-Order Linear Systems).

For the first-order linear system (2), if all entries of $\mathbf{P}(t)$ and $\mathbf{g}(t)$ are continuous on an interval I , then for any $t_0 \in I$ and $\mathbf{y}_0 \in \mathbb{R}^n$, there exists a unique solution $\mathbf{y}(t)$ to the IVP

$$\begin{cases} \mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t) \\ \mathbf{y}(t_0) = \mathbf{y}_0 \end{cases} \quad \text{for } t \in I.$$